

IHC - Practical Approach to Data Mining and Analytics

Modular Assignment No. 6

Module 6: Text Analysis - Image Analytics and Text Analytics | Duration: 4 hours

Problem Statement

Design and implement a comprehensive unstructured Data Analytics solution that demonstrates mastery of image analytics and text analytics techniques. Students will build an integrated platform that processes, analyses, and extracts insights from visual and textual data using advanced computer vision, natural language processing, and deep learning methodologies to solve real-world business problems.

Dataset Sources

Students must work with **at least 4 datasets** covering both image and text domains:

Image Analytics Datasets

Computer Vision Classics:

- **CIFAR-10/100** (<https://www.cs.toronto.edu/~kriz/cifar.html>) - Object classification
- **ImageNet** (<http://www.image-net.org/>) - Large-scale image recognition
- **COCO Dataset** (<https://cocodataset.org/>) - Object detection and segmentation
- **Open Images** (<https://storage.googleapis.com/openimages/web/index.html>)
- **Fashion-MNIST** (<https://github.com/zalandoresearch/fashion-mnist>)

Medical Imaging:

- **NIH Chest X-rays** (<https://www.nih.gov/news-events/news-releases>)
- **ISIC Skin Cancer** (<https://www.isic-archive.com/>)
- **Diabetic Retinopathy** (Kaggle) - Eye disease detection
- **Brain MRI** (Kaggle) - Tumor detection datasets

Satellite and Geospatial:

- **Sentinel-2** (ESA) - Earth observation imagery
- **Landsat** (USGS) - Land use classification
- **Planet Labs** - High-resolution satellite imagery
- **Urban Atlas** - City planning and development

Industry-Specific Images:

- **Manufacturing Defects** - Quality control datasets
- **Retail Product Images** - E-Commerce visual search

- **Traffic and Transportation** - Vehicle detection, road analysis
- **Agricultural Imagery** - Crop monitoring, precision farming

Text Analytics Datasets

Social Media and Reviews:

- **Twitter API** - Real-time sentiment and trend analysis
- **Reddit API** - Community discussion analysis
- **Amazon Product Reviews** - E-Commerce sentiment analysis
- **Yelp Reviews** - Restaurant and business analytics
- **TripAdvisor Reviews** - Tourism and hospitality insights

News and Media:

- **Reuters-21578** - News categorization and topic modeling
- **20 Newsgroups** - Multi-category text classification
- **All The News (Kaggle)** - News article analysis
- **BBC News** - News classification and summarization
- **CNN/DailyMail** - Text summarization datasets

Academic and Scientific:

- **arXiv Papers** - Scientific literature analysis
- **PubMed Abstracts** - Medical literature mining
- **Wikipedia Dumps** - Knowledge extraction and topic modeling
- **Google Books Ngrams** - Language evolution analysis
- **Patent Datasets** - Innovation and technology trends

Business and Legal:

- **SEC Filings (EDGAR)** - Financial document analysis
- **Legal Case Documents** - Legal text mining
- **Customer Support Tickets** - Issue classification and resolution
- **Survey Responses** - Qualitative data analysis

Multimodal Datasets

Vision-Language Tasks:

- **MS-COCO Captions** - Image captioning and description
- **Visual Question Answering (VQA)** - Image-text reasoning
- **Flickr30k** - Image-text alignment and retrieval
- **Conceptual Captions** - Large-scale image-text pairs

Social Media Multimodal:

- **Instagram Posts** - Visual and textual content analysis
 - **YouTube Videos** - Video, audio, and text analytics
 - **TikTok Data** - Short-form video and text analysis
 - **Pinterest Pins** - Visual search and recommendation
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What is Expected

Students must demonstrate comprehensive understanding of unstructured Data Analytics through practical implementation of Computer Vision techniques, Natural Language Processing(NLP) methods, and integrated multimodal analysis. The solution should showcase feature extraction, pattern recognition, sentiment analysis, and business insight generation while maintaining scalability and real-world applicability.

Rubrics and Evaluation Criteria

1. Image Analytics Implementation (35 points)

- **Excellent (32-35):** Advanced Computer Vision techniques with Deep Learning and custom architectures
- **Good (25-31):** Good implementation with appropriate preprocessing and feature extraction
- **Satisfactory (18-24):** Basic image processing with standard techniques
- **Needs Improvement (0-17):** Inadequate image analysis or poor technical implementation

Assessment Components

- **Image Preprocessing:** Resizing, normalization, augmentation, noise reduction
- **Feature Extraction:** SIFT, SURF, HOG, deep features from CNNs
- **Classification Models:** CNN architectures (LeNet, AlexNet, ResNet, EfficientNet)
- **Object Detection:** YOLO, R-CNN, SSD for object localization and identification
- **Image Segmentation:** Semantic and instance segmentation techniques
- **Transfer Learning:** Pre-trained model fine-tuning and feature adaptation

2. Text Analytics Implementation (35 points)

- **Excellent (32-35):** Advanced NLP techniques with transformer models and custom solutions
- **Good (25-31):** Good text processing with appropriate Machine Learning approaches
- **Satisfactory (18-24):** Basic text analysis with standard NLP techniques
- **Needs Improvement (0-17):** Inadequate text processing or poor analytical depth

Assessment Components

- **Text Preprocessing:** Tokenization, stemming, lemmatization, stop word removal
- **Feature Engineering:** TF-IDF, word embeddings (Word2Vec, GloVe), contextual embeddings

- **Sentiment Analysis:** Rule-based, machine learning, and Deep Learning approaches
- **Topic Modeling:** LDA, NMF, BERTopic for thematic analysis
- **Named Entity Recognition:** Person, organization, location extraction and classification
- **Text Classification:** Multi-class and multi-label document classification

3. Advanced Techniques and Deep Learning (30 points)

- **Excellent (27-30):** State-of-the-art Deep Learning models with custom architectures
- **Good (21-26):** Good Deep Learning implementation with appropriate frameworks
- **Satisfactory (15-20):** Basic Deep Learning with standard models
- **Needs Improvement (0-14):** Limited or incorrect Deep Learning implementation

Assessment Components

- **Convolutional Neural Networks:** Custom CNN architectures for image analysis
 - **Recurrent Neural Networks:** LSTM, GRU for sequential text analysis
 - **Transformer Models:** BERT, GPT, T5 for advanced text understanding
 - **Generative Models:** GANs for image generation, GPT for text generation
 - **Multimodal Models:** Vision-language models for integrated analysis
 - **Transfer Learning:** Pre-trained model adaptation and fine-tuning strategies
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Deliverables

1. Integrated Analytics Platform (50%)

- **Image Processing Pipeline:** Complete Computer Vision workflow with multiple techniques
- **Text Analysis Framework:** Comprehensive NLP pipeline with advanced features
- **Multimodal Integration:** Combined image-text analysis capabilities
- **API Development:** RESTful services for image and text analytics

2. Advanced Models and Algorithms (30%)

- **Custom Deep Learning Models:** Original architectures for specific domain problems
- **Ensemble Methods:** Combining multiple models for improved performance
- **Real-time Processing:** Streaming analytics for live image and text data
- **Performance Optimization:** GPU acceleration and distributed computing

3. Business Application Portfolio (20%)

- **Industry Use Cases:** Real-world applications across different domains
 - **Interactive Dashboards:** Visual interfaces for image and text analytics
 - **Insight Generation:** Automated reporting and pattern discovery
 - **Scalability Demonstration:** Performance testing with large datasets
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Technical Requirements

Core Technical Stack

- **Python Frameworks:** OpenCV, PIL, scikit-image for image processing
- **Deep Learning:** TensorFlow, PyTorch, Keras for neural networks
- **NLP Libraries:** NLTK, spaCy, Hugging Face Transformers
- **Computer Vision:** torchvision, tensorflow-hub, detectron2
- **Visualization:** matplotlib, seaborn, plotly for analytics visualization

Advanced Requirements

- **GPU Computing:** CUDA-enabled processing for Deep Learning
- **Cloud Services:** AWS Rekognition, Google Vision API, Azure Cognitive Services
- **Big Data Processing:** Spark MLlib for large-scale text/image processing
- **Real-time Streaming:** Apache Kafka, Redis for live data processing
- **Model Serving:** Flask, FastAPI, TensorFlow Serving for model deployment

Specialized Tools

- **Image Processing:** ImageIO, scikit-image, Mahotas for advanced operations
- **Text Processing:** Gensim, TextBlob, polyglot for multilingual analysis
- **Annotation Tools:** LabelImg, VGG Image Annotator for dataset preparation
- **Model Management:** MLflow, Weights and Biases for experiment tracking

Advanced Options (Bonus Points)

- **Edge Computing:** TensorFlow Lite, ONNX for mobile/edge deployment
- **Distributed Training:** Horovod, PyTorch Distributed for large-scale training
- **AutoML:** AutoKeras, Neural Architecture Search for automated model design
- **Federated Learning:** Privacy-preserving distributed model training
- **Explainable AI:** LIME, SHAP, GradCAM for model interpretability

Timeline and Milestones

Hour 1: Image Analytics Implementation

- Image dataset loading and preprocessing pipeline
- Feature extraction using traditional and Deep Learning methods
- Image classification and object detection implementation
- Transfer learning and model fine-tuning

Hour 2: Text Analytics Development

- Text preprocessing and feature engineering pipeline
- Sentiment analysis and topic modeling implementation

- Named entity recognition and text classification
- Advanced NLP with transformer models

Hour 3: Multimodal Integration and Advanced Techniques

- Image-text multimodal analysis implementation
- Custom deep learning architecture development
- Ensemble methods and model optimization
- Real-time processing capabilities

Hour 4: Deployment and Business Applications

- API development and model serving setup
- Performance testing and scalability analysis
- Business application development and demonstration
- Documentation and presentation preparation

Specific Implementation Requirements

Image Analytics Module

```
# Example structure - students must implement
class ImageAnalyzer:
    def __init__(self):
        self.preprocessors = {}
        self.extractors = {}
        self.classifiers = {}
        self.detectors = {}

    def preprocess_image(self, image, operations=['resize', 'normalize']):
        # Comprehensive image preprocessing
        pass

    def extract_features(self, image, method='cnn'):
        # Traditional and deep learning feature extraction
        pass

    def classify_image(self, image, model_type='resnet'):
        # Image classification with multiple architectures
        pass

    def detect_objects(self, image, model='yolo'):
        # Object detection and localization
        pass

    def segment_image(self, image, method='semantic'):
        # Image segmentation techniques
        pass
```

Text Analytics Module

```
# Example structure - students must implement
class TextAnalyzer:
```

```

def __init__(self):
    self.preprocessors = {}
    self.embeddings = {}
    self.classifiers = {}
    self.generators = {}

def preprocess_text(self, text, operations=['tokenize', 'lemmatize']):
    # Comprehensive text preprocessing
    pass

def extract_features(self, text, method='tfidf'):
    # Traditional and neural text representations
    pass

def analyze_sentiment(self, text, model='transformer'):
    # Multi-approach sentiment analysis
    pass

def extract_topics(self, documents, method='lda'):
    # Topic modeling and thematic analysis
    pass

def extract_entities(self, text, model='spacy'):
    # Named entity recognition and classification
    pass

```

Multimodal Analytics Module

Example structure - students must implement

```
class MultimodalAnalyzer:
```

```

    def __init__(self):
        self.image_analyzer = ImageAnalyzer()
        self.text_analyzer = TextAnalyzer()
        self.fusion_models = {}

    def analyze_image_text_pair(self, image, text, task='similarity'):
        # Integrated image-text analysis
        pass

    def generate_captions(self, image, model='transformer'):
        # Image captioning and description
        pass

    def visual_question_answering(self, image, question):
        # VQA using multimodal models
        pass

    def cross_modal_retrieval(self, query, modality='text'):
        # Cross-modal search and retrieval
        pass

```

Evaluation Scenarios

Scenario 1: Social Media Content Analytics

- **Image Analytics:** Brand logo detection, product recognition, content moderation

- **Text Analytics:** Sentiment analysis, hashtag analysis, influencer identification
- **Multimodal:** Image-caption consistency, viral content prediction
- **Expected Output:** Social media monitoring platform with real-time analytics

Scenario 2: Healthcare Document and Image Analysis

- **Image Analytics:** Medical image diagnosis, X-ray analysis, skin lesion detection
- **Text Analytics:** Clinical note processing, medical literature mining, drug interaction analysis
- **Multimodal:** Radiology report generation, medical image captioning
- **Expected Output:** Healthcare analytics system with diagnostic support

Scenario 3: E-Commerce Product Intelligence

- **Image Analytics:** Product categorization, visual search, quality inspection
- **Text Analytics:** Review sentiment, product description optimization, customer feedback
- **Multimodal:** Product recommendation, visual-textual product matching
- **Expected Output:** E-Commerce intelligence platform with visual and text analytics

Scenario 4: News and Media Content Analysis

- **Image Analytics:** Photo journalism analysis, fake image detection, visual trend analysis
- **Text Analytics:** Article categorization, bias detection, fact-checking, summarization
- **Multimodal:** News story verification, multimedia content analysis
- **Expected Output:** Media analytics platform with content verification capabilities

Performance Benchmarks

Image Analytics Performance

- **Classification Accuracy:** >90% on CIFAR-10, >75% on ImageNet
- **Processing Speed:** >100 images/second on standard hardware
- **Object Detection:** mAP >0.5 on COCO dataset
- **Memory Efficiency:** Process 4K images with <8GB RAM

Text Analytics Performance

- **Sentiment Accuracy:** >85% on standard sentiment datasets
- **Processing Speed:** >1000 documents/second for basic NLP
- **Topic Coherence:** >0.5 coherence score for topic models
- **NER F1-Score:** >0.85 on standard NER benchmarks

Deep Learning Performance

- **Training Convergence:** <50 epochs for standard datasets
- **Model Size:** <100MB for deployment-ready models

- **Inference Latency:** <100ms for real-time applications
- **GPU Utilization:** >80% GPU memory utilization during training

Scalability Metrics

- **Concurrent Processing:** Handle 1000+ simultaneous requests
 - **Batch Processing:** Process 10K+ images or documents in <1 hour
 - **Real-time Streaming:** <500ms end-to-end latency for live data
 - **Storage Efficiency:** <1GB storage per 10K processed items
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Innovation Challenges (Bonus)

1. Multimodal Fusion Architecture (20 bonus points)

- Develop novel architectures for image-text fusion
- Create attention mechanisms for cross-modal understanding
- Implement zero-shot learning for unseen image-text combinations
- Build unified embedding spaces for images and text

2. Domain Adaptation and Transfer Learning (15 bonus points)

- Develop domain adaptation techniques for specialized datasets
- Create few-shot learning systems for rapid domain switching
- Implement progressive transfer learning strategies
- Build domain-agnostic feature extractors

3. Explainable AI for Unstructured Data (15 bonus points)

- Implement attention visualization for text and image models
- Create saliency maps for image classification decisions
- Develop text explanation systems for NLP models
- Build interactive explanation interfaces for end users

4. Edge and Mobile Deployment (10 bonus points)

- Optimise models for mobile and edge devices
 - Implement quantization and pruning techniques
 - Create offline-capable analytics applications
 - Build battery-efficient processing pipelines
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Advanced Topics (Optional)

1. Generative Models

- **GANs for Images:** StyleGAN, CycleGAN for image generation and transformation
- **Text Generation:** GPT-based models for content creation
- **Multimodal Generation:** Image-to-text and text-to-image generation
- **Style Transfer:** Neural style transfer and domain adaptation

2. Self-Supervised Learning

- **Contrastive Learning:** SimCLR, SwAV for image representation learning
- **Masked Language Modeling:** BERT-style pretraining for text
- **Multimodal Self-Supervision:** CLIP-style training for vision-language tasks
- **Augmentation Strategies:** Advanced data augmentation techniques

3. Federated and Privacy-Preserving Analytics

- **Federated Learning:** Distributed training across multiple clients
 - **Differential Privacy:** Privacy-preserving analytics techniques
 - **Secure Multi-party Computation:** Collaborative analytics without data sharing
 - **Homomorphic Encryption:** Computation on encrypted data
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Industry Integration Requirements

1. Computer Vision APIs

- **Google Vision API:** OCR, object detection, face recognition integration
- **AWS Rekognition:** Image analysis, video analysis, content moderation
- **Azure Computer Vision:** Image analysis, spatial analysis, read API
- **IBM Watson Visual Recognition:** Custom model training and deployment

2. NLP Cloud Services

- **Google Natural Language AI:** Entity analysis, sentiment, syntax analysis
- **AWS Comprehend:** Key phrase extraction, language detection, custom entities
- **Azure Text Analytics:** Sentiment analysis, key phrase extraction, language detection
- **IBM Watson Natural Language Understanding:** Emotion analysis, concepts, categories

3. MLOps and Model Management

- **Kubeflow:** Machine learning workflows on Kubernetes
- **MLflow:** Experiment tracking, model versioning, deployment
- **Weights and Biases:** Experiment tracking, hyperparameter tuning
- **TensorBoard:** Visualization and monitoring of training progress

Submission Requirements

1. Complete Analytics Platform

- **Source Code:** Well-documented, modular implementation for both domains
- **Model Artifacts:** Trained models with evaluation metrics and documentation
- **API Documentation:** RESTful API with OpenAPI specification
- **Deployment Guide:** Container deployment and scaling instructions

2. Technical Portfolio

- **Algorithm Comparison:** Comprehensive comparison of different approaches
- **Performance Analysis:** Benchmarking results and optimization strategies
- **Scalability Study:** Load testing and distributed processing analysis
- **Innovation Documentation:** Novel approaches and custom implementations

3. Business Application Suite

- **Use Case Demonstrations:** Real-world applications across multiple domains
- **Interactive Interfaces:** User-friendly web interfaces for analytics
- **Automated Reporting:** Insight generation and pattern discovery systems
- **Integration Examples:** API usage examples and third-party integrations

Academic Integrity and Professional Standards

Data Ethics and Privacy

- Proper handling of personal data in images and text
- Bias detection and mitigation in computer vision and NLP models
- Ethical considerations for content analysis and monitoring
- Transparency in automated decision-making systems

Model Reproducibility

- Comprehensive documentation of training procedures
- Version control for datasets, models, and experiments
- Reproducible research practices with proper seed setting
- Open science considerations and result sharing

Technical Excellence

- Industry-standard coding practices and documentation
- Comprehensive testing including unit, integration, and performance tests
- Security considerations for model deployment and data handling

- Accessibility compliance for user interfaces and visualizations
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Support Resources

Technical Documentation

- Computer Vision algorithm references and implementation guides
- Natural Language Processing technique documentation and tutorials
- Deep Learning architecture guides and best practices
- Multimodal learning research papers and implementation examples

Development Environment

- Pre-configured development environments with GPU support
- Cloud-based training platforms (Google Colab, AWS SageMaker)
- Dataset preparation and annotation tool guides
- Model deployment and serving platform documentation

Performance Optimization

- GPU optimization techniques and memory management
- Model compression and quantization strategies
- Distributed training and inference optimization
- Edge deployment and mobile optimization guides

Total Points: 100

Bonus Points: Up to 60

Passing Grade: 70 points

Time Limit: 4 hours